L1 Perception

(a) Relative distance

Input image



Q. Which is closest: the person, table, or snack bag?

A. snack bag B. table

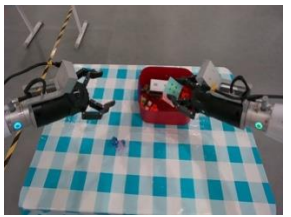
C. person

L2 Embodied Semantics

(b) Action between frames

Earlier frame *Later frame*





Q. What happened between these two frames?

B. Pick up the rubik's cube on the table with the right arm

L3 Planning & Reasoning

(c) Action prediction/selection

Frame 1



Frame 3



Frame 6



Frame 8 current



Q. Goal: refill coffee bean hopper. What should happen next?

C. Lift the left hopper lid off